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A Two-Stage Modeling Approach to Risk Prediction with High-Dimensional Two-Phase Data

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PROJECT PAGE & CONTACT

Perelman SCHOOL OF MEDICINE UNIVERSITY of PENNSYLVANIA

CLINICAL MOTIVATION

Patient	Age	Sex	SBP	...	Pain score	Fatigue score	...
XXX	55	M	96		Missing		
XXX	43	M	114		2	4	
XXX	34	F	92		Missing		
...							

Complete EHR Data — Incomplete PRO Data

Phase I: high-dimensional EHR predictors $\mathbf{X}$ and outcome Y are observed for all patients.
Phase II: granular external predictors $\mathbf{Z}$ (surveys, biobanks, wearables) are observed only for a subset of patients.
Limitations of common approaches:

- complete-case analysis loses information and may not generalize;
- direct imputation may be unstable when phase-II availability is low or $\mathbf{Z}$ weakly related to $\mathbf{X}$.

Goal: Build and evaluate a risk model for binary outcomes using all phase-I data without directly imputing missing external variables.

TWO-STAGE FRAMEWORK

Key assumption: missing at random, $P(R=1|Y,\mathbf{X},\mathbf{Z}) = P(R=1|Y,\mathbf{X})$.

STAGE I
Learn initial risk from all N patients

Flexible model: $Y \sim \mathbf{X}$
K-fold sample splitting procedure
Risk score: $l(\mathbf{X}) = \text{logit}\{P^I(Y=1|\mathbf{X})\}$

STAGE II
Recalibrate + Revise using observed Z

Primary model: $Y \sim l(\mathbf{X}) + \mathbf{Z}$
Pseudo-score for coefficient estimation

Primary Model

$$\text{logit}\{P(Y=1|l(\mathbf{X}),\mathbf{Z};\beta)\} = \beta_0 + \beta_1 l(\mathbf{X}) + \beta_2^T\mathbf{Z}$$

Coefficient Estimation

- **Selection correction:** incorporate the estimated log availability-odds ratio as an offset in the pseudo-score equation *(Breslow and Cain, 1988)*.
- **Availability model:** use a B-spline basis of the stage-I residual $p^{l}_{1-y}(\mathbf{X})$, with optional low-dimensional $\mathbf{X}_r$.

$$\text{logit}\{P(R=1|Y=y,\mathbf{X})\} = \boldsymbol{\alpha}_y^T\mathbf{X}_r + \boldsymbol{\delta}_y^T\mathbf{B}\{p^{l}_{1-y}(\mathbf{X})\}$$

AUC Estimation

- Estimate the **weighted empirical distribution** of $(l(\mathbf{X}), Z)$:

$$\hat{F}(l,\mathbf{z}) = \frac{1}{N}\sum_i \frac{I\{R_i=1, \hat{l}_i \le l, \mathbf{Z}_i \le \mathbf{z}\}}{\hat{\pi}_i},$$

$$\hat{\pi}_i = \sum_y \hat{P}(R_i=1 \mid Y_i=y,\mathbf{X}_i)\,\hat{P}(Y_i=y \mid \hat{l}_i,\mathbf{Z}_i).$$

where $\hat{\pi}_i$ is the estimated probability that individual i has phase-II data.

- Under adequate stage-II calibration, obtain **plug-in estimates** of TPR, FPR, and AUC.

SIMULATION STUDY

Data generation

- $N = 10{,}000$
- 300 EHR predictors (10 signals) + 3 external predictors
- $(\mathbf{X},\mathbf{Z}) \sim N(\mathbf{0},\boldsymbol{\Sigma})$, $\Sigma_{ij} = \rho^{|i-j|}$, $\rho \in \{0, 0.3, 0.5\}$

Outcome & Missingness

$$\text{logit}\{P(Y=1|\mathbf{X},\mathbf{Z})\} = \phi_0 + \boldsymbol{\phi_1}\boldsymbol{X} + \boldsymbol{\phi_2}\boldsymbol{Z}; \quad \text{logit}\{P(R=1|\mathbf{X})\} = \tau_0 + \boldsymbol{\tau_1}\frac{\sum_{j=1}^{q} X_j}{\sqrt{q}}$$

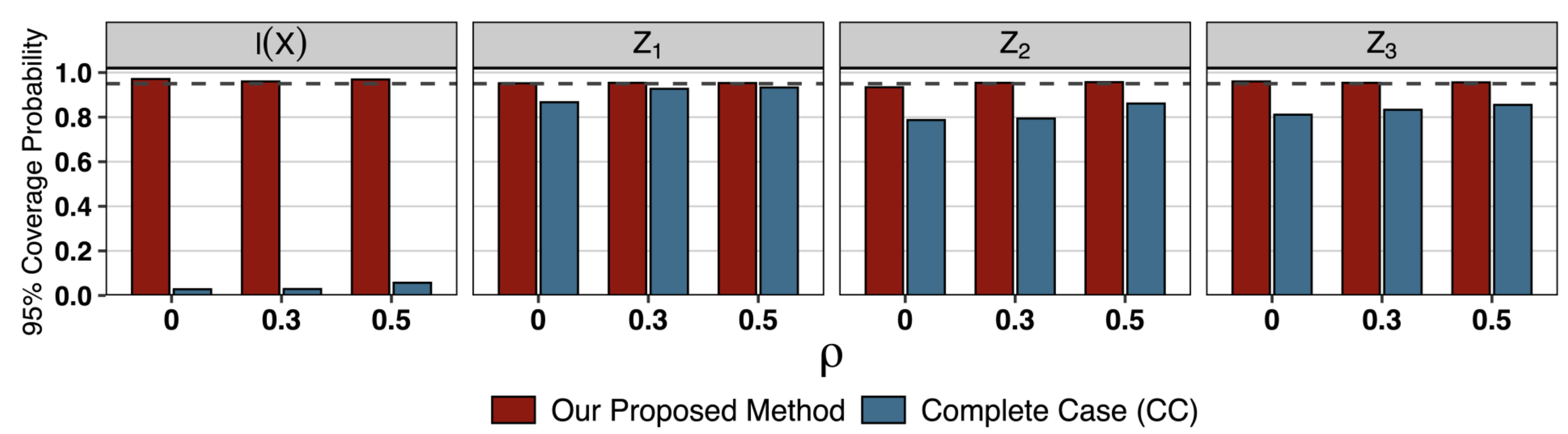
Implementation

- Stage I: adaptive lasso
- 1,000 simulation replicates

Comparators

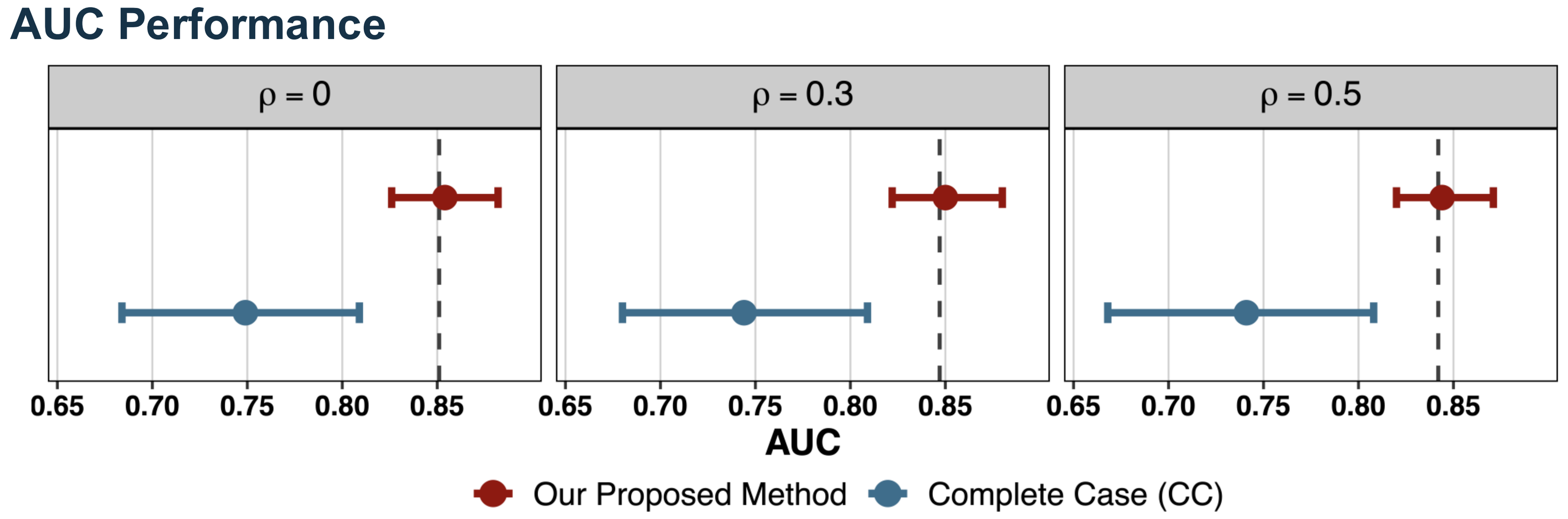
- Complete-data two-stage benchmark
- Complete-case (CC)

Coefficient Inference



95% coverage probabilities under $P(R=1)=0.05$, evaluated relative to the complete-data benchmark. The dashed line indicates the nominal 95% coverage level.

AUC Performance



Mean AUCs (points) and empirical 95% intervals (horizontal bars) under $(P(R=1)=0.05)$; dashed lines denote the complete-data AUCs.

Simulation takeaway. Even at only 5% phase-II availability, the proposed coefficient and AUC estimates remained close to the complete-data benchmark.

REAL-WORLD ONCOLOGY APPLICATION

Predict 180-day mortality by enriching structured EHR data with patient-reported outcomes (PROs).

8,555	5.7%	204	4	55%
Patients	180-day deaths	EHR predictors	PROs	PRO availability

Two-Stage I modeled PRO availability using a smooth function of the stage-I residual.
Two-Stage II additionally included the 10 EHR predictors most strongly associated with PRO availability.

Log-odds coefficients and AUC (95% CI)

Predictor / metric	Two-Stage I	Two-Stage II	Complete Case
Intercept	-0.695 (-1.624, 0.235)	-0.675 (-1.601, 0.252)	-0.115 (-1.050, 0.820)
$\hat{l}(\mathbf{X})$	0.713 (0.643, 0.782)	0.707 (0.636, 0.779)	1.000 (0.857, 1.143)
Performance Status	0.379 (0.208, 0.550)	0.383 (0.213, 0.553)	0.306 (0.126, 0.487)
Nausea Frequency	0.119 (-0.021, 0.259)	0.117 (-0.024, 0.258)	0.128 (-0.020, 0.276)
Numbness & Tingling	-0.194 (-0.342, -0.046)	-0.197 (-0.345, -0.050)	-0.185 (-0.341, -0.029)
Quality of Life	0.216 (0.045, 0.388)	0.221 (0.051, 0.392)	0.154 (-0.020, 0.329)
AUC	**0.820 (0.796, 0.845)**	**0.821 (0.796, 0.846)**	**0.867 (0.818, 0.915)**

Robust availability adjustment: Two-Stage I and II were nearly identical, suggesting the smooth residual term captured the relevant response pattern.

Complete-case results may reflect a different patient distribution: coefficient shifts and the higher CC AUC are consistent with differences in X between PRO responders and nonresponders.

CONCLUSIONS

- Risk score $l(\mathbf{X})$ **summarizes high-dimensional phase-I predictors** while **leveraging information from the full cohort.**
- The framework supports **flexible choices of stage-I prediction models**.
- Pseudo-score estimation **accounts for distributional differences** between the observed phase-II sample and the target population, yielding **reliable estimates even at 5% phase-II availability** in simulations.

Selected references:
Breslow & Cain (1988). Logistic regression for two-stage case-control data. Biometrika, 75:11–20.
Hasler et al. (2024). A semiparametric method for risk prediction using integrated electronic health record data. Ann. Appl. Stat., 18:3318–3337.

CODE & SUPPLEMENT: github.com/chenyubi-code/TwoPhaseHighD